British Airways Data Science Virtual Experience Program Report

Task : Predicting Customer Booking Behaviour

Project Overview

The objective of this project was to develop a machine learning model capable of predicting whether a customer would complete a booking with British Airways. By identifying the factors that influence booking completion, the company can improve customer targeting, optimize marketing campaigns, and increase conversion rates.

Dataset Description

The dataset contained approximately 50,000 customer booking records with 13 input features and one target variable.

Target Variable

- **booking_complete**
  - 0 = Booking Not Completed
  - 1 = Booking Completed

Key Features

- Number of Passengers
- Sales Channel
- Trip Type
- Purchase Lead Time
- Length of Stay
- Flight Hour
- Flight Day
- Route
- Booking Origin
- Extra Baggage Purchase
- Preferred Seat Purchase
- In-flight Meal Purchase
- Flight Duration

Data Preparation

The dataset was cleaned and prepared before model development.
The following preprocessing steps were performed:

- Loaded and inspected the dataset
- Checked for missing values
- Removed duplicate records
- Converted categorical variables into numerical format using Label Encoding
- Defined feature and target variables
- Split the dataset into training and testing sets

Machine Learning Model

A **Random Forest Classifier** was selected because it performs well on classification problems, handles mixed feature types effectively, and provides feature importance scores that help explain model predictions.

Model Configuration

- Algorithm: Random Forest Classifier
- Train-Test Split: 80%-20%
- Cross Validation: 5-Fold Cross Validation

Model Performance
Accuracy Score
85.24%
Cross Validation Score
49.64%
The model achieved strong overall accuracy. However, the dataset was imbalanced, with significantly fewer completed bookings than incomplete bookings. As a result, future improvements could include techniques such as class balancing, SMOTE, or class weighting to improve prediction performance for completed bookings.

Feature Importance Analysis
The model identified the following features as the most influential predictors of booking completion:

1. Purchase Lead
2. Route
3. Flight Hour
4. Length of Stay
5. Booking Origin
6. Flight Day
7. Flight Duration

These variables contributed the most to the model's predictive capability.

Key Findings

- Purchase lead time was the strongest predictor of booking completion.
- Route selection significantly influenced customer purchasing behavior.
- Flight departure timing affected booking outcomes.
- Length of stay and booking origin contributed to customer decision-making.
- Flight duration also played an important role in booking completion.
- Customers purchasing additional services demonstrated stronger booking intent.

Business Recommendations
Based on the findings, the following recommendations are proposed:
1. Personalized Marketing
Target customers with characteristics associated with higher booking completion probability.
2. Optimize Promotional Campaigns
Design route-specific and time-based promotional strategies to increase conversion rates.
3. Encourage Ancillary Purchases
Promote additional services such as baggage, meals, and seat selection to improve customer engagement.
4. Leverage Predictive Analytics
Deploy machine learning models to identify high-value customers and support data-driven marketing decisions.

Conclusion
The Random Forest model successfully identified important factors influencing customer booking completion. The analysis revealed that purchase lead time, route characteristics, flight timing, and customer booking patterns significantly impact booking behavior. These insights can help British Airways improve customer targeting, enhance marketing effectiveness, and support data-driven business decisions.

Prepared By
Nasimuddin Ansari
B.Tech in [AI & DS] at SRMU Lucknow